How to Load and Run a Program

Hiwonder ROS Robot Controller V1.2 · without an ST-Link · MTR 520 2DOF Drone Rig

> **You need:** STM32CubeIDE, STM32CubeProgrammer, Tera Term, and one USB-C cable. The board has two serial USB-C ports: **UART1** (left) is for **loading**, **UART2** (middle) is for **watching** the printed output. With one cable, move it between them.

1 Build the program (STM32CubeIDE)

Click the **hammer icon** (or Ctrl+B). The Console must end with **0 errors**.
The program is the **.elf** file in the project's **Debug** folder, e.g. **`Debug/ImuBoardTest.elf`**. To find it on disk: right-click the .elf → **Show In → System Explorer**.

2 Load it (STM32CubeProgrammer, cable in UART1)

Plug the cable into **UART1** (left USB-C). In CubeProgrammer, change the top-right dropdown from ST-LINK to **UART** and set:

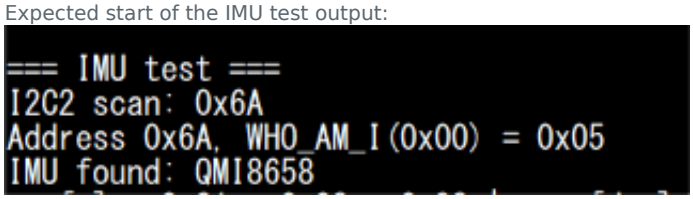
Setting	Value
Port	COM port of the board (refresh if the list is empty)
Baudrate	115200
Parity	Even
DTR	**High** (required: this puts the board into loading mode)

Then: **Connect → Erasing & Programming** (download icon, left) → **Browse** to the .elf → **Start Programming** → wait for "download complete" → **Disconnect**.

3 Run and watch it (Tera Term, cable in UART2)

Move the cable to **UART2** (middle USB-C). The board powers up and runs the program (LED blinks).
Tera Term: **File → New connection → Serial** → pick the COM port → **Setup → Serial port → Speed 115200**. Press the board's **RST** button to see the start-up lines.

Expected start of the IMU test output:



Troubleshooting

Problem	Fix
CubeProgrammer won't connect	Check **DTR = High**, Parity Even, cable in **UART1**. Close Tera Term if it has the same COM port open. Press RST and try again.
Don't know which COM port	Windows **Device Manager → Ports (COM & LPT)**. Unplug the cable and see which entry disappears. UART1 and UART2 show up as different COM numbers.
Programmed, but LED doesn't blink	Click **Disconnect** in CubeProgrammer, then press **RST** (or unplug and replug).
Tera Term shows nothing or garbage	Cable must be in **UART2**, speed **115200**, correct COM port. Press RST.
Decimal numbers print as blank	CubeIDE: Project → Properties → C/C++ Build → Settings → MCU Settings → tick **Use float with printf from newlib-nano**, then rebuild.

Every code change: build → cable in UART1 → load → cable to UART2 → watch.